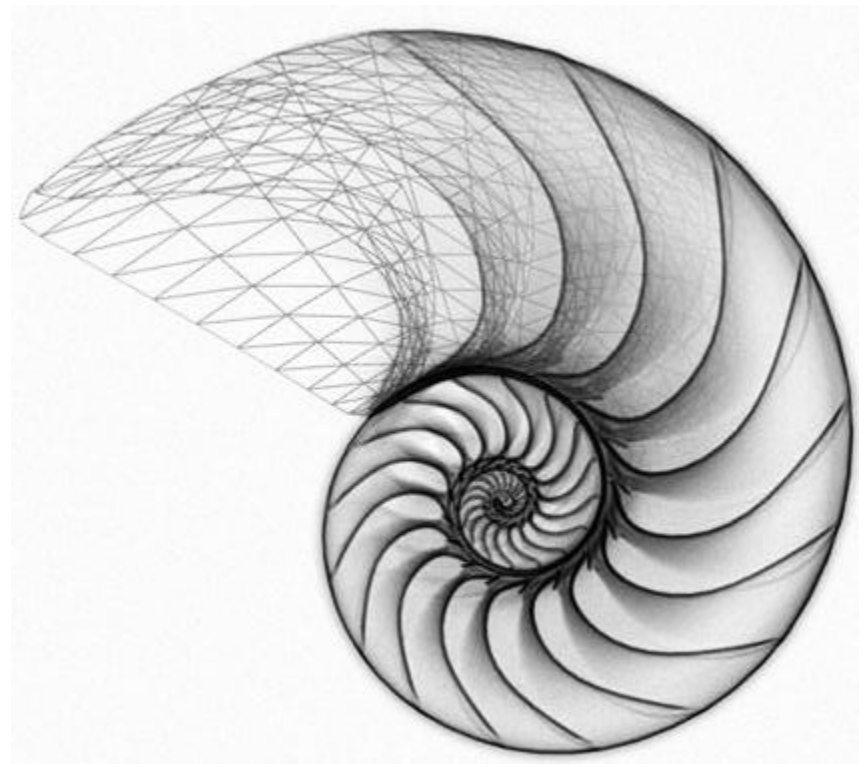




Lecture 19

Sorting

CSE225: Data Structures and Algorithms

Radix Sort

- Take the least significant digit of the values to sort.
- Group the values based on that digit, but otherwise keep the original order of values.
- Repeat the grouping process with each more significant digit.

Radix Sort

- How do we get the i^{th} significant digit of a number in constant time?

Radix Sort

- How do we get the i^{th} significant digit of a number in constant time?

8

4

7

3

4

2

9

1

Radix Sort

- How do we get the i^{th} significant digit of a number in constant time?

8

4

7

3

4

2

9

1

- Position 1: $8749 \% 10 = 9$

Radix Sort

- How do we get the i^{th} significant digit of a number in constant time?

8

4

7

3

4

2

9

1

- Position 1: $8749 \% 10 = 9$
- Position 2: $(8749 / 10) \% 10 = 4$

Radix Sort

- How do we get the i^{th} significant digit of a number in constant time?

8

4

7

3

4

2

9

1

- Position 1: $8749 \% 10 = 9$
- Position 2: $(8749 / 10) \% 10 = 4$
- Position 3: $(8749 / 100) \% 10 = 7$

Radix Sort

- How do we get the i^{th} significant digit of a number in constant time?

8

4

7

3

4

2

9

1

- Position 1: $8749 \% 10 = 9$
- Position 2: $(8749 / 10) \% 10 = 4$
- Position 3: $(8749 / 100) \% 10 = 7$
- Position 4: $(8749 / 1000) \% 10 = 8$

Radix Sort

- How do we get the i^{th} significant digit of a number in constant time?

8

4

7

3

4

2

9

1

- Position 1: $(8749 / 1) \% 10 = 9$
- Position 2: $(8749 / 10) \% 10 = 4$
- Position 3: $(8749 / 100) \% 10 = 7$
- Position 4: $(8749 / 1000) \% 10 = 8$

Radix Sort

- How do we get the i^{th} significant digit of a number in constant time?

8

4

7

3

4

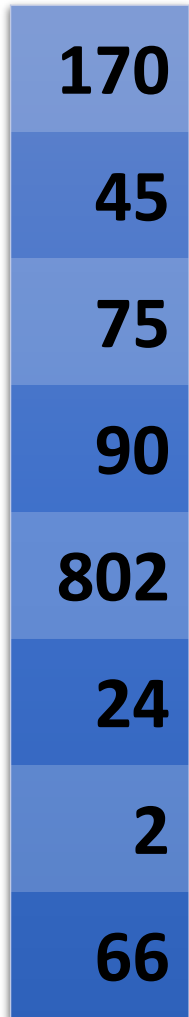
2

9

1

- Position 1: $(8749 / 1) \% 10 = 9$
- Position 2: $(8749 / 10) \% 10 = 4$
- Position 3: $(8749 / 100) \% 10 = 7$
- Position 4: $(8749 / 1000) \% 10 = 8$
- **Position i: $(value / 10^{i-1}) \% 10$**

Radix Sort



Radix Sort

170	1 7 0
45	0 4 5
75	0 7 5
90	0 9 0
802	8 0 2
24	0 2 4
2	0 0 2
66	0 6 6

Radix Sort

170	1	7	0
45	0	4	5
75	0	7	5
90	0	9	0
802	8	0	2
24	0	2	4
2	0	0	2
66	0	6	6

Radix Sort

170	1	7	0
45	0	4	5
75	0	7	5
90	0	9	0
802	8	0	2
24	0	2	4
2	0	0	2
66	0	6	6

Array of queues						
[0]						
[1]						
[2]						
[3]						
[4]						
[5]						
[6]						
[7]						
[8]						
[9]						

Radix Sort

170	1	7	0
45	0	4	5
75	0	7	5
90	0	9	0
802	8	0	2
24	0	2	4
2	0	0	2
66	0	6	6

Array of queues						
[0]	170					
[1]						
[2]						
[3]						
[4]						
[5]						
[6]						
[7]						
[8]						
[9]						

Radix Sort

170	1	7	0
45	0	4	5
75	0	7	5
90	0	9	0
802	8	0	2
24	0	2	4
2	0	0	2
66	0	6	6

Array of queues						
[0]	170					
[1]						
[2]						
[3]						
[4]						
[5]	45					
[6]						
[7]						
[8]						
[9]						

Radix Sort

170	1	7	0
45	0	4	5
75	0	7	5
90	0	9	0
802	8	0	2
24	0	2	4
2	0	0	2
66	0	6	6

Array of queues						
[0]	170					
[1]						
[2]						
[3]						
[4]						
[5]	45	75				
[6]						
[7]						
[8]						
[9]						

Radix Sort

170	1	7	0
45	0	4	5
75	0	7	5
90	0	9	0
802	8	0	2
24	0	2	4
2	0	0	2
66	0	6	6

Array of queues						
[0]	170	90				
[1]						
[2]						
[3]						
[4]						
[5]	45	75				
[6]						
[7]						
[8]						
[9]						

Radix Sort

170	1	7	0
45	0	4	5
75	0	7	5
90	0	9	0
802	8	0	2
24	0	2	4
2	0	0	2
66	0	6	6

Array of queues						
[0]	170	90				
[1]						
[2]	802					
[3]						
[4]						
[5]	45	75				
[6]						
[7]						
[8]						
[9]						

Radix Sort

170	1	7	0
45	0	4	5
75	0	7	5
90	0	9	0
802	8	0	2
24	0	2	4
2	0	0	2
66	0	6	6

Array of queues						
[0]	170	90				
[1]						
[2]	802					
[3]						
[4]	24					
[5]	45	75				
[6]						
[7]						
[8]						
[9]						

Radix Sort

170	1	7	0
45	0	4	5
75	0	7	5
90	0	9	0
802	8	0	2
24	0	2	4
2	0	0	2
66	0	6	6

Array of queues						
[0]	170	90				
[1]						
[2]	802	2				
[3]						
[4]	24					
[5]	45	75				
[6]						
[7]						
[8]						
[9]						

Radix Sort

170	1	7	0
45	0	4	5
75	0	7	5
90	0	9	0
802	8	0	2
24	0	2	4
2	0	0	2
66	0	6	6

Array of queues						
[0]	170	90				
[1]						
[2]	802	2				
[3]						
[4]	24					
[5]	45	75				
[6]	66					
[7]						
[8]						
[9]						

Radix Sort

170	1	7	0
90	0	9	0
802	8	0	2
2	0	0	2
24	0	2	4
45	0	4	5
75	0	7	5
66	0	6	6

Array of queues						
[0]	170	90				
[1]						
[2]	802	2				
[3]						
[4]	24					
[5]	45	75				
[6]	66					
[7]						
[8]						
[9]						

Radix Sort

170	1	7	0
90	0	9	0
802	8	0	2
2	0	0	2
24	0	2	4
45	0	4	5
75	0	7	5
66	0	6	6

Radix Sort

170	1	7	0
90	0	9	0
802	8	0	2
2	0	0	2
24	0	2	4
45	0	4	5
75	0	7	5
66	0	6	6

Radix Sort

170	1	7	0
90	0	9	0
802	8	0	2
2	0	0	2
24	0	2	4
45	0	4	5
75	0	7	5
66	0	6	6

Array of queues						
[0]						
[1]						
[2]						
[3]						
[4]						
[5]						
[6]						
[7]						
[8]						
[9]						

Radix Sort

170	1	7	0
90	0	9	0
802	8	0	2
2	0	0	2
24	0	2	4
45	0	4	5
75	0	7	5
66	0	6	6

Array of queues						
[0]	802	2				
[1]						
[2]	24					
[3]						
[4]	45					
[5]						
[6]	66					
[7]	170	75				
[8]						
[9]	90					

Radix Sort

802	8	0	2
2	0	0	2
24	0	2	4
45	0	4	5
66	0	6	6
170	1	7	0
75	0	7	5
90	0	9	0

Array of queues						
[0]	802	2				
[1]						
[2]	24					
[3]						
[4]	45					
[5]						
[6]	66					
[7]	170	75				
[8]						
[9]	90					

Radix Sort

802	8	0	2
2	0	0	2
24	0	2	4
45	0	4	5
66	0	6	6
170	1	7	0
75	0	7	5
90	0	9	0

Radix Sort

802	8	0	2
2	0	0	2
24	0	2	4
45	0	4	5
66	0	6	6
170	1	7	0
75	0	7	5
90	0	9	0

Radix Sort

802	8	0	2
2	0	0	2
24	0	2	4
45	0	4	5
66	0	6	6
170	1	7	0
75	0	7	5
90	0	9	0

Array of queues						
[0]						
[1]						
[2]						
[3]						
[4]						
[5]						
[6]						
[7]						
[8]						
[9]						

Radix Sort

802	8	0	2
2	0	0	2
24	0	2	4
45	0	4	5
66	0	6	6
170	1	7	0
75	0	7	5
90	0	9	0

Array of queues						
[0]	2	24	45	66	75	90
[1]	170					
[2]						
[3]						
[4]						
[5]						
[6]						
[7]						
[8]	802					
[9]						

Radix Sort

2	0	0	2
24	0	2	4
45	0	4	5
66	0	6	6
75	0	7	5
90	0	9	0
170	1	7	0
802	8	0	2

Array of queues						
[0]	2	24	45	66	75	90
[1]	170					
[2]						
[3]						
[4]						
[5]						
[6]						
[7]						
[8]	802					
[9]						

Radix Sort



Radix Sort



Time complexity:

$$O(d \times n)$$

Where, d = number of digits

Counting Sort

```
1      CountingSort(A, B, k)
2          for i=1 to k
3              C[i]= 0;
4          for j=1 to n
5              C[A[j]] += 1;
6          for i=2 to k
7              C[i] = C[i] + C[i-1];
8          for j=n downto 1
9              C[A[j]] -= 1;
10         B[C[A[j]]] = A[j];
```

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	0	0	0	0	0

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	1	0	0	0	0

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	1	1	0	0	0

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	1	1	0	1	0

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	1	2	0	1	0

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	1	2	1	1	0

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	1	2	1	1	1

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	1	2	1	2	1

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	2	2	1	2	1

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	2	3	1	2	1

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	1	2	3	1	2	1

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	2	2	3	1	2	1

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	2	2	3	1	3	1

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	2	2	3	1	4	1

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	2	3	3	1	4	1

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	2	3	3	2	4	1

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	2	3	3	2	5	1

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	2	3	3	2	5	1

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	2	5	3	2	5	1

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	2	5	8	2	5	1

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	2	5	8	10	5	1

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	2	5	8	10	15	1

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	2	5	8	10	15	16

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	2	5	8	10	15	16

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	2	5	8	10	14	16

B

[illegible]

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	2	5	8	9	14	16

B

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value										3					4	

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	2	4	8	9	14	16

B

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value					1					3					4	

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	2	4	8	9	13	16

B

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value					1					3				4	4	

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	2	4	8	9	12	16

B

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value					1					3			4	4	4	

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	1	4	8	9	12	16

B

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value		0			1					3			4	4	4	

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	4	8	9	12	16

B

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	0	0			1					3			4	4	4	

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	4	7	9	12	16

B

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	0	0			1			2		3			4	4	4	

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	3	7	9	12	16

B

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	0	0		1	1			2		3			4	4	4	

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	3	7	9	11	16

B

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	0	0		1	1			2		3		4	4	4	4	

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	3	7	9	11	15

B

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	0	0		1	1			2		3		4	4	4	4	5

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	3	7	8	11	15

B

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	0	0		1	1			2	3	3		4	4	4	4	5

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	3	6	8	11	15

B

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	0	0		1	1		2	2	3	3		4	4	4	4	5

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	3	6	8	10	15

B

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	0	0		1	1		2	2	3	3	4	4	4	4	4	5

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	3	5	8	10	15

B

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	0	0		1	1	2	2	2	3	3	4	4	4	4	4	5

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	2	5	8	10	15

B

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	0	0	1	1	1	2	2	2	3	3	4	4	4	4	4	5

Counting Sort

A

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	1	2	4	2	3	5	4	1	2	0	0	4	4	1	3	4

C

index	0	1	2	3	4	5
value	0	2	5	8	10	15

B

index	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
value	0	0	1	1	1	2	2	2	3	3	4	4	4	4	4	5